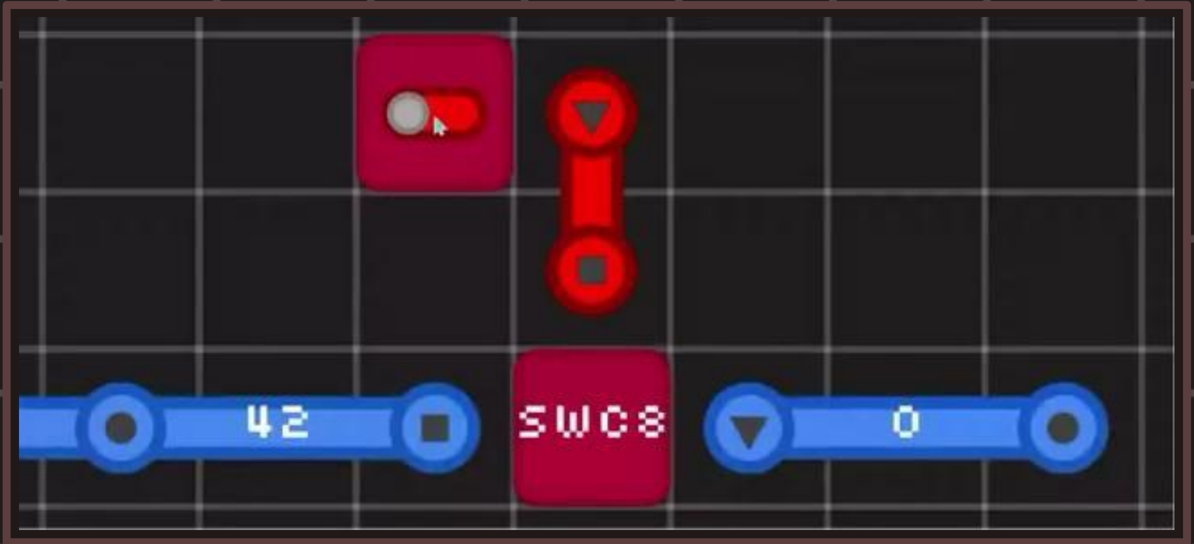
Digital Logic
Day 6

uarch.space

uarch.space/slides/day6.pdf

Switch component

The Switch component allows enables or disables an 8-bit wire.

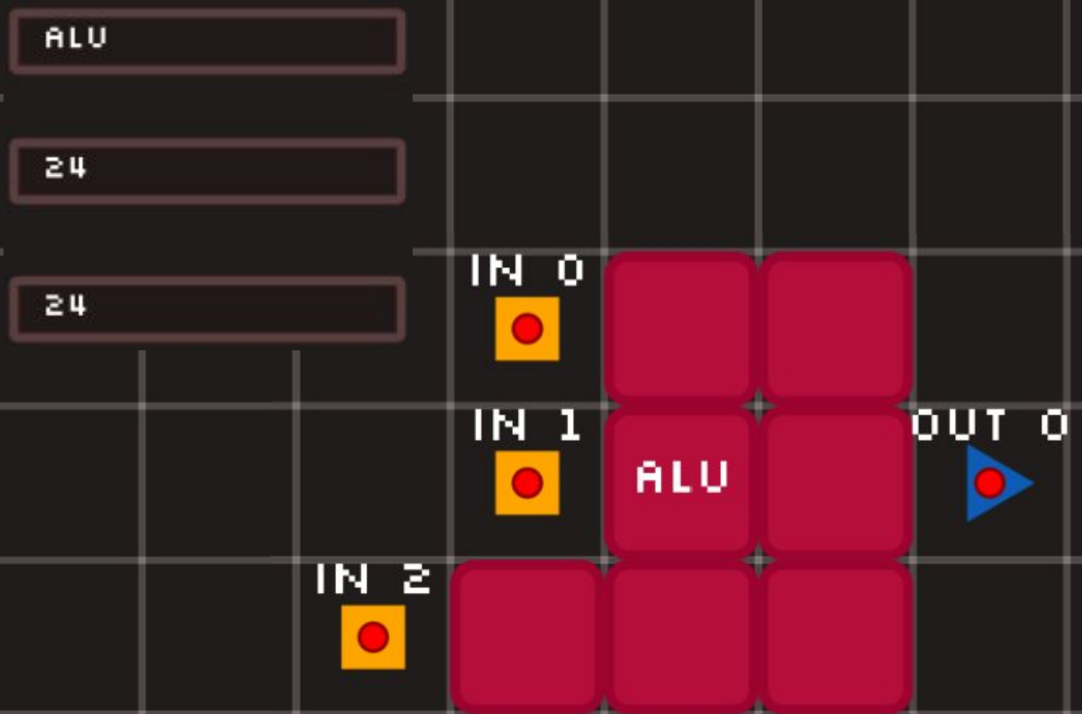


ALU (Arithmetic Logic Unit)

The ALU is like a calculator.

Like on a calculator, you can select what operation to perform and what your operands are.

All inputs are 8 bit.

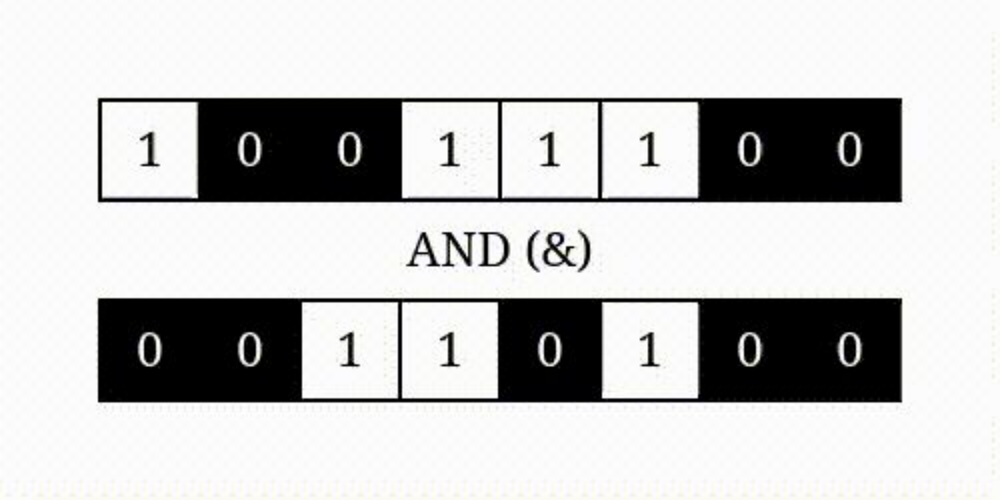


Bitwise Operators

Bitwise operations are like the existing logic gates you know, except the operate on 8 bits at a same time.

Only columns where:

(1 AND 1) outputs 1



Bitwise AND

https://realpython.com/python-bitwise-operators/

Bitwise Operators

Bitmasked Right Shift (>>)

Bitmasked Left Shift (<<)

NOT (~)

(<<) is equivalent to multiplying by two.

(>>) is equivalent to dividing by two.

https://realpython.com/python-bitwise-operators/

ALU (Arithmetic Logic Unit)

These are the operations that we will use.

IDENTITY just outputs the first operand without any calculation.

Some operations ignore the second operand entirely.

000	ADD	Adds the two operands
001	AND	Bitwise ANDs the two operands
010	OR	Bitwise ORs the two operands
011	XOR	Bitwise XORs the two operands
100	NOT	Inverts the first operand
101	SHR	Bit Shift operand 1 right 1 bit
110	SHL	Bit Shift operand 1 left 1 bit
111	IDENTITY	Output operand 1

Solutions:

uarch.space/solutions/day6.pdf